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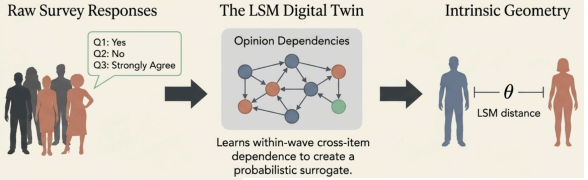
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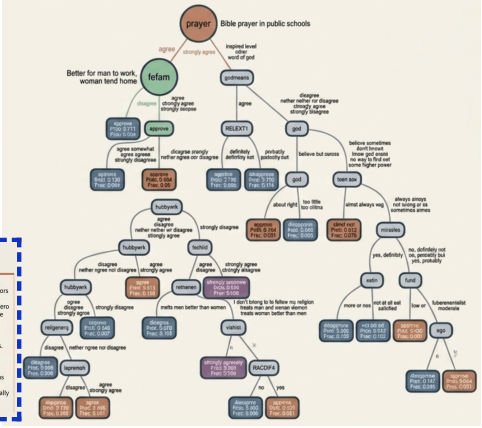
Geometry of Opinions

Building Data-driven Digital Twins from Social Surveys

DARPA-EA-25-02-05-MAGICS-PA-025
HR0011-26-3-E016

The Large Science Model (LSM) Framework





The Model

For each survey item, the LSM estimates a conditional distribution using a CIT.

$$\Phi_P^i : \prod_{j \neq i} \Sigma_j \to \mathcal{D}(\Sigma_i)$$

The Recursive Forest: Variables used as features in one CIT (like fefam) have their own conditional models. This creates a mutually constrained probabilistic structure over partially observed vectors.

The Intrinsic Metric (θ)

$$\theta_{P,Q}(x,y) \triangleq \mathbb{E}_i \left[J^{1/2} \left(\Phi_P^i(x_{-i}), \Phi_Q^i(y_{-i}) \right) \right]$$

The LSM distance between two partially observed response vectors.

The expected value across all survey items in the instrument.

The square root of the Jensen-Shannon divergence (ensuring distances sum linearly along sequences).

The conditional response distribution induced by the contemporaneous twin.

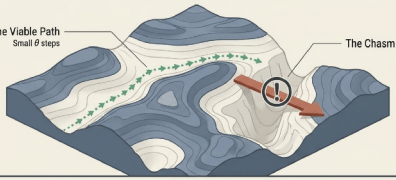
Intrinsic Discrepancy

Key Insight: Distinct vectors may have a distance of zero if they induce identical conditional response distributions.

Socially irrelevant perturbations are mathematically ignored.

Inference

Opinion Geometry & Transition Probabilities

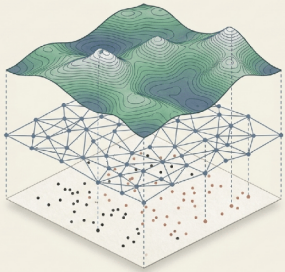


$$\theta_P(x,y) \propto -\log \Pr_P(x \to y)$$

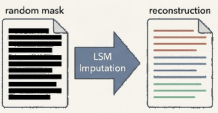
- **Small θ (The Viable Path):** Represents statistically feasible perturbations under the contemporaneous dependency structure.
- **Large θ (The Chasm):** Direct transitions between distant worldviews are exponentially suppressed. Distance equals the cost of deformation.

The Zeitgeist as a Falsifiable Object

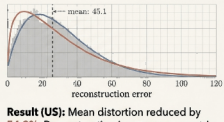
- The Zeitgeist is no longer a metaphor. It is a measurable, time-specific constraint structure that dictates the feasible worldviews of a population.
- By learning high-dimensional dependencies directly from data, the LSM replaces descriptive indices with an operational, generative geometry.
- Societal change appears not as movement along a predefined axis, but as the physical deformation of the constraint fabric itself.



Result I: Out-of-Sample Reconstruction



The Test: Mask 80% of responses for held-out participants. Impute the masked coordinates using the contemporaneous LSM.



Result (US): Mean distortion reduced by 54.9%. Reconstruction improves geometric fidelity for >90% of GSS respondents.

Result (EU): Mean post-reconstruction distortion drops to 39.1%. Error is reduced for >99.5% of EU respondents.

The learned geometry captures genuine, predictive dependency patterns across individuals and decades.

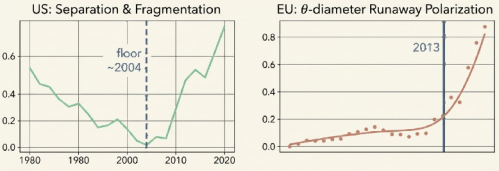
Scale of Validation

General Social Survey (GSS)		Eurobarometer	
1972–2021 Timeframe	65,784 Respondents	235 Surveys through July 2022	3,980,244 Participants
31,670,949 Item-level responses	6,209 Unique survey items	> 500 Questions per survey on average	

Validation

Application

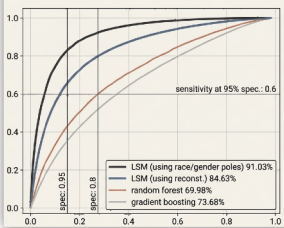
Interpretation I: The Deformation of Opinion Space



Because the geometry is population-indexed, we can track fixed reference poles over time. Polarization is measured as the actual *deformation* of feasible opinion configurations within the society.

Insight

Result II: Predicting Real-World Behavior



Target: 2016 US Presidential Voting choice (PRES16)

AUC > 91%
LSM using Race/Gender Poles

AUC > 84%
LSM using Direct Reconstruction

AUC < 70%
Random Forest Baseline

Sidebar Takeaway: Predictability is not explained by demographics alone. LSM-derived opinion predictors discover non-obvious cross-domain linkages that vastly outperform standard ML demographic models.